

M.Tech Project Zeroth Review

INITIAL PLANNING STAGE – PRE-IMPLEMENTATION

This presentation outlines two proposed AI/ML projects at the zeroth review stage. No implementation has been completed; all architectures, datasets, and workflows represent planned approaches subject to revision.

Project 1

Data Narrator: NL-to-SQL BI Copilot with Auto-Insights

Category: Generative AI | Domain: Analytics / Enterprise BI

Project 2

RAG-Based Intelligent Defect Detection and Knowledge Assistance for Smart Manufacturing with Azure Cloud Deployment and Interactive Visualization

Domain: Smart Manufacturing / Industrial AI

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Proposed Projects
Both at initial planning stage

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Implementation Complete
No models trained or deployed yet

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AI/ML Paradigms
Generative AI + RAG + Classical ML

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Cloud Targets
Docker (Project 1) + Microsoft Azure (Project 2)

Data Narrator: NL-to-SQL BI Copilot with Auto-Insights

PROJECT 1

GENERATIVE AI | ANALYTICS / ENTERPRISE BI

Problem Statement

Business users in enterprise organizations may depend on analysts to generate reports because of SQL skill gaps and analytics-team backlogs. This can delay access to business information and increase dependence on technical teams.

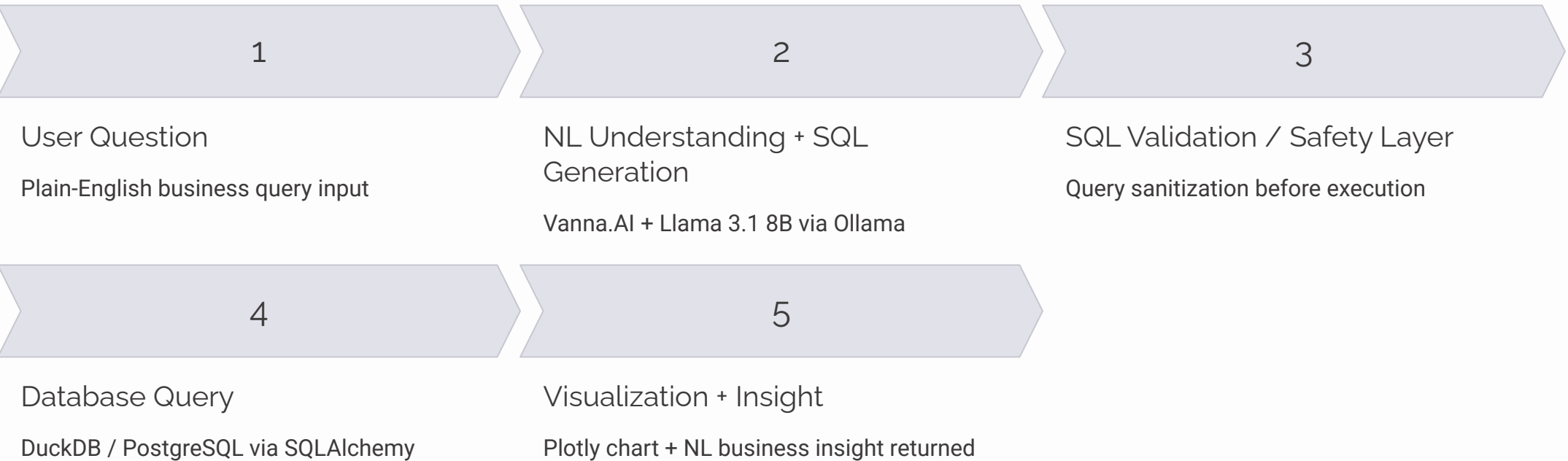
Proposed Solution

Develop a natural-language Business Intelligence assistant that allows non-technical users to ask questions about business data in plain English. The system will convert the question into validated SQL, execute the query against a structured database, automatically select an appropriate visualization, and generate a short natural-language business insight.

Objectives

- 01
- Convert natural-language questions into SQL queries.
- 02
- Execute validated SQL queries against a structured database.
- 03
- Automatically select an appropriate chart type.
- 04
- Generate concise natural-language business insights.
- 05
- Support conversational follow-up questions using previous query context.
- 06
- Provide a simple chat-based BI interface.

Proposed End-to-End Workflow



Data Narrator — Dataset & Technical Architecture

PROJECT 1 — ARCHITECTURE

Proposed Dataset: Tableau Sample Superstore Sales

Source: Tableau Public Sample Data | **Type:** Structured transactional/business dataset

Selection Rationale: Contains business dimensions and numerical measures suitable for testing natural-language business queries, aggregations, comparisons, trend analysis, and automated visualizations.

Attribute	Analytical Purpose
Order ID	Identifies individual orders
Order Date	Supports time and trend analysis
Customer / Segment	Customer-level and segment comparison
Region	Geographical analysis
Category / Sub-Category	Product-category and detailed product analysis
Product Name	Product-level analysis
Sales / Quantity	Revenue and sales-volume analysis
Discount / Profit	Discount and profitability analysis

Example Natural-Language Queries

"Which region generated the highest sales?"

"Show the monthly sales trend."

"Which category has the highest profit?"

"Compare profit between customer segments."

Proposed Technical Stack

Component	Technology
Programming	Python
NL-to-SQL	Vanna.AI
LLM	Ollama + Llama 3.1 8B
Database	DuckDB / PostgreSQL
SQL Execution	SQLAlchemy
Visualization	Plotly
Backend	FastAPI + Pydantic
Frontend	Streamlit
Deployment	Docker

Expected Outcome

A conversational BI assistant that will allow non-technical users to ask questions in natural language and receive data-driven answers, SQL-generated results, appropriate visualizations, automated business insights, and context-aware follow-up responses.

RAG-Based Intelligent Defect Detection and Knowledge Assistance for Smart Manufacturing

PROJECT 2

SMART MANUFACTURING / INDUSTRIAL AI

Problem Statement

Manufacturing systems generate sensor data related to process and product conditions. Complex and overlapping defect patterns can make defect analysis challenging. Engineers also need quick access to manufacturing SOPs, quality manuals, maintenance procedures, and troubleshooting information.

Proposed Solution

Develop an intelligent manufacturing system combining Machine Learning and Retrieval-Augmented Generation (RAG). The ML component will classify manufacturing components according to defect severity. The RAG component will retrieve relevant engineering knowledge from a domain-specific knowledge base and use an LLM to generate explanations and corrective recommendations.

Target Defect Classes

No Defect	Minor Defect
Moderate Defect	Critical Defect

Objectives

- 01
- Generate realistic synthetic manufacturing sensor data.
- 02
- Classify manufacturing components according to defect severity.
- 03
- Build a domain-specific manufacturing knowledge base.
- 04
- Retrieve relevant technical documents using semantic search.
- 05
- Generate AI-based explanations and corrective recommendations.
- 06
- Analyze and visualize defect patterns.
- 07
- Develop a scalable cloud-based architecture using Microsoft Azure.

Proposed End-to-End Workflow



Smart Manufacturing Defect Detection — Dataset & Technical Architecture

PROJECT 2 — ARCHITECTURE

Proposed Dataset: Synthetic Manufacturing Sensor Data

Proposed Size: 10,000–50,000 manufacturing records. The dataset will be synthetically generated to simulate realistic manufacturing conditions, including variability, noise, overlapping defect patterns, and outlier failures.

Input Attribute	Represents
Surface Roughness Index	Surface-quality characteristics
Vibration Level	Machine/process vibration conditions
Temperature Deviation	Deviation from expected thermal conditions
Dimensional Deviation	Variation in component dimensions
Material Stress Indicator	Material/process stress conditions

Proposed ML Models & Evaluation Metrics

Random Forest — Baseline Classification Model

XGBoost — Comparison Model

Accuracy | Precision | Recall | F1-Score | Confusion Matrix

RAG Knowledge Base Contents

- Manufacturing SOP documents
- Quality inspection manuals
- Defect handling guidelines
- Maintenance procedures
- Engineering troubleshooting documents

Full Technical Stack

Component	Technology
Embedding Model	Sentence Transformers
Vector Database	FAISS / ChromaDB
RAG Framework	LangChain
LLM Integration	OpenAI API / Azure OpenAI
Data Processing	Pandas, NumPy
Visualization	Plotly, Matplotlib
Frontend / Backend	Streamlit / FastAPI / Flask
Database	PostgreSQL / Azure SQL
Big Data	Apache Spark / PySpark
Cloud & Storage	Microsoft Azure + Azure Blob Storage
Deployment	Docker

Expected Outcome

A smart manufacturing assistant that will predict defect category, provide prediction confidence, retrieve relevant engineering knowledge, generate an AI-based defect explanation, recommend corrective engineering actions, and display defect and model information through an interactive dashboard.